

UCI Online Retail Customer Segmentation

Final report

Anish Ranjan

2026-05-04

1 Abstract

We build an unsupervised customer-segmentation pipeline on one year of UCI Online Retail transactions [7] (~4,300 customers, ~391k positive line items, UK gift retailer, Dec 2010 – Dec 2011). The pipeline reduces seven engineered RFM-and-behavioural features to three principal components (~80% variance retained) and partitions customers with k-means at $k = 3$ [1], chosen by the gap statistic [2] and validated by silhouette width [3], adjusted Rand index against Ward hierarchical clustering [4], and bootstrap Jaccard stability [5]. The headline finding is the **revenue–customer asymmetry**: cluster A (~28% of customers) holds ~75% of net revenue, defining where retention spend should go and which customers can be treated as low-margin volume. Next steps are time-aware features, a soft-assignment Gaussian mixture model benchmark, and a controlled marketing experiment per segment.

2 Overview

Problem statement. A retailer that treats every customer the same overspends on low-value buyers and underspends on the few that drive revenue. The business question *who do we protect, who do we grow, who do we accept as low-margin?* needs a small, defensible vocabulary tied to behaviour rather than gut feel.

Proposed methodology. Per-customer feature engineering -> column-wise standardisation -> PCA for dimensionality reduction -> k-means on PC scores with k selected by gap-statistic SE-max -> multi-criterion validation (silhouette, ARI vs Ward, bootstrap Jaccard) -> A/B/C re-labelling by descending median net spend -> persisted scoring artefact for new customers.

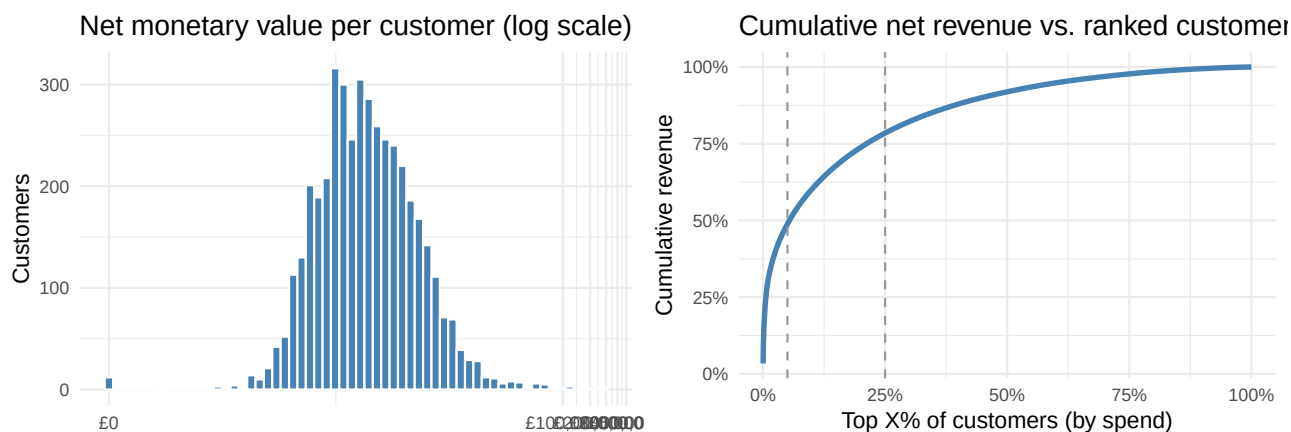
3 Data Processing

Cleaning (`src/python/data_prep.py`) drops missing-customer rows, deduplicates, removes non-product stock codes (postage, bank charges), **separates cancellation invoices into a sibling table** rather than discarding them (so net monetary can be computed honestly), and retains positive-quantity / positive-price lines in the main table. After cleaning: ~391k positive line items across ~4,334 customers plus a ~9k cancellations table.

Issues and adjustments. (i) `AvgUnitPrice` is computed quantity-weighted rather than as a line-level mean, which would overweight tiny low-priced items. (ii) `AvgUnitPrice` is winsorised at its 99th percentile so a few extreme prices do not dominate PC2/PC3. (iii) Right-skewed features get $\log(1 + x)$; `NetMonetary`, which can be slightly negative, gets the sign-preserving variant $\text{sgn}(x) \log(1 + |x|)$. (iv) StockCode 23843 is a single bulk order whose product-level revenue dwarfs others; product-mix tables are reported both with and without it.

4 Data Analysis

Per-customer features: Recency, Frequency, GrossMonetary, NetMonetary, TotalQuantity, UniqueProducts, AvgOrderValue, AvgUnitPrice (winsorised), TenureDays, plus profiling-only fields (Country, ReturnRate, ReturnValueRate, SinglePurchase). Two patterns in the EDA shape every modelling decision:

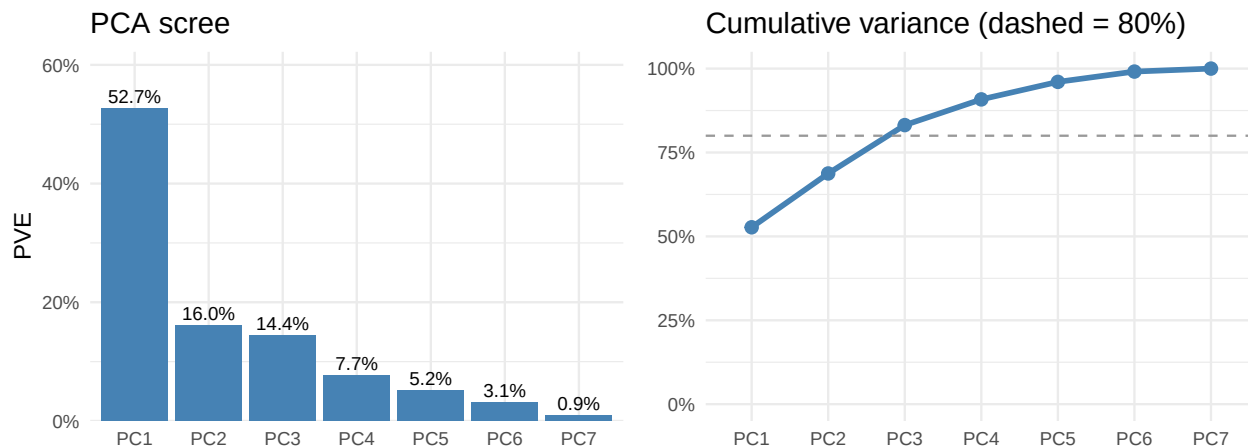


Heavy right-skew on spend. Net monetary spans four orders of magnitude distance-based methods are dominated by the upper tail without a log transform. **Revenue is highly concentrated.** The top 5% of customers account for ~half of net revenue and the top 25% for the great majority. This is the empirical shape the segmentation must respect: separating the high-value tail from the low-value mass is the minimum criterion for a useful partition.

5 Model Training

Feature engineering for the model matrix. Seven features survive EDA-driven pruning (collinear blocks collapsed, profiling-only fields held back): `log_Recency`, `log_Frequency`, `log_NetMonetary` (sign-preserving variant, since net monetary can be slightly negative), `log_UniqueProducts`, `log_AvgOrderValue`, `AvgUnitPrice_winsor`, and `TenureDays`. Columns are centred and scaled to unit variance.

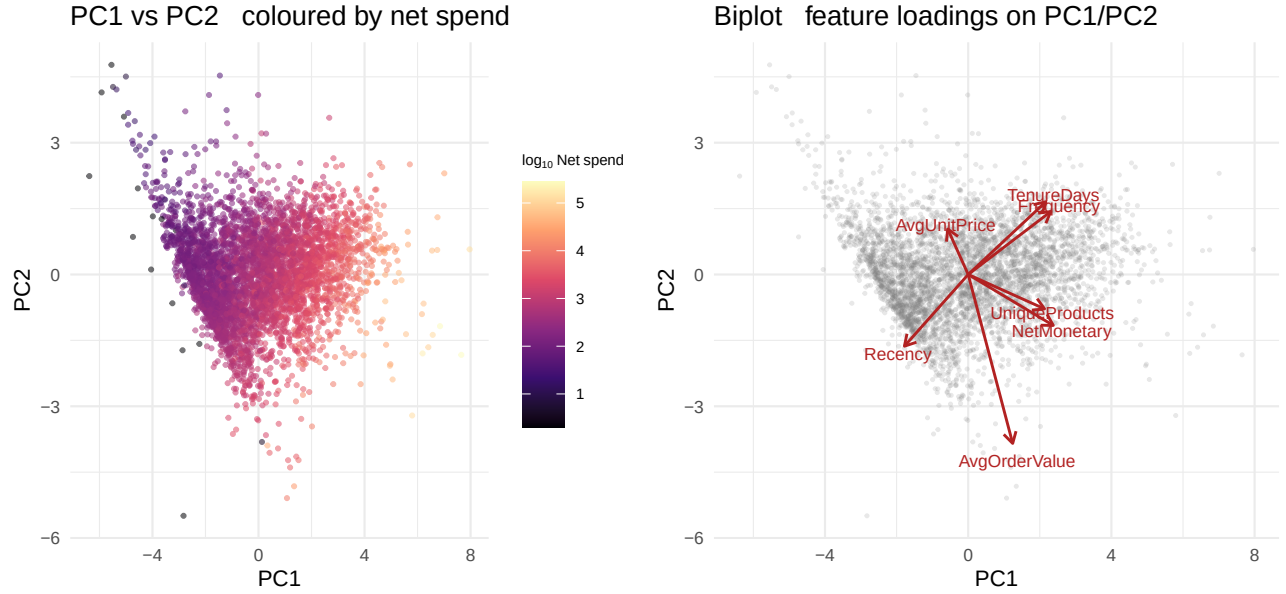
Evaluation metrics for k selection. (i) Within-cluster sum of squares (elbow heuristic). (ii) Mean silhouette width [3]: $s(i)$ contrasts the within-cluster tightness $a(i)$ against the nearest-other-cluster distance $b(i)$. (iii) Gap statistic [2] with the SE-max rule, biased toward parsimony. We sweep $k \in \{2, \dots, 10\}$ with `nstart` = 50 and a fixed seed.



Model selection. Three principal components retain $\sim 80\%$ of the variance and the elbow flattens after PC3 adding a fourth buys little variance and costs interpretability. PC1 is the engagement / customer- value axis (positive on F , M^{net} , U , T ; negative on R); PC2 is an order-size vs tenure contrast (negative on A , P); PC3 is the residual price-point axis. For k , silhouette peaks at $k = 2$ and remains competitive at $k = 3$; the gap statistic SE-max rule selects $k = 3$. We adopt $k = 3$ as the canonical segmentation; $k = 2$ is retained as a sensitivity model.

6 Model Validation

Three independent validations confirm the partition is real, not an artefact of the algorithm.



(i) **PC1 captures spend.** The colour gradient on the left panel moves cleanly along PC1 the principal axis recovered from variance maximisation aligns with the business-meaningful spend axis. The biplot confirms: **Frequency**, **NetMonetary**, **UniqueProducts**, and (negatively) **Recency** all load along PC1, while **AvgOrderValue** and **AvgUnitPrice** load on PC2 and define wholesale-vs-retail.

(ii) **Bootstrap Jaccard stability** [5]. Resampling customers with replacement and refitting, every k-means cluster recovers above the 0.75 stability threshold ($\sim 0.87+$); Ward at the same k produces one stable cluster and two borderline ones. (iii) **ARI** [4] between k-means and Ward is in the moderate range the cluster *cores* are shared, the *boundaries* differ. K-means is promoted as the canonical label source on the grounds of stability.

Risks and biases. PC1's larger variance weights it more heavily under Euclidean distance, so the segmentation is partially driven by that axis; the raw-vs-scaled-PC ARI sensitivity reports how much that matters in practice. The high-value tail pulls cluster A's centroid; re-fitting with the top 1% removed is a recommended robustness check.

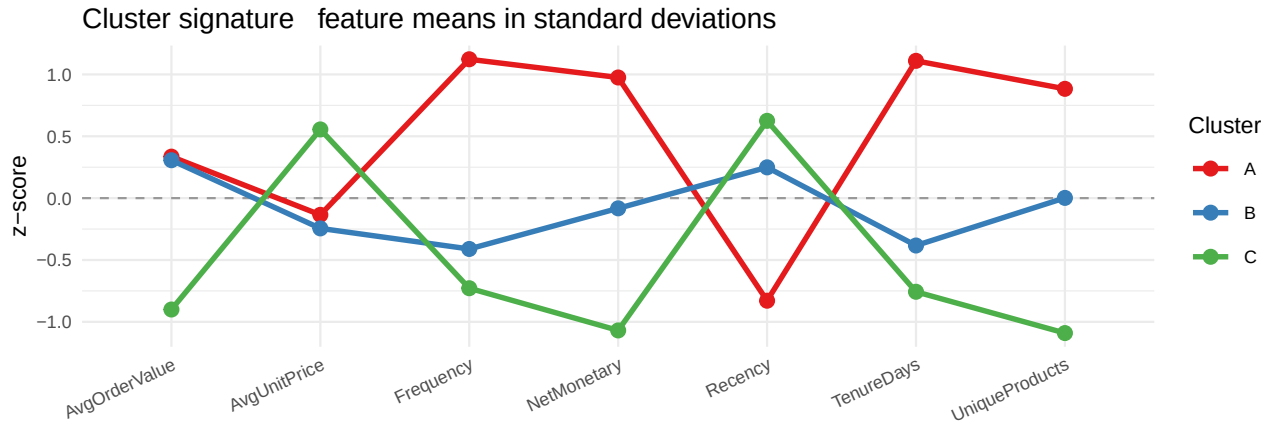
7 Model Performance

Table 1: Per-cluster headline numbers (live from the pipeline output)

Cluster	Persona	Customers	% cust.	% rev.	Median net spend	Med. orders	Med. recency
A	Loyal high-value	1,396	32.2%	79.5%	£2,265	6	16 d
B	Developing mid-value	1,804	41.6%	17.5%	£600	2	70 d
C	Lapsed one-time	1,134	26.2%	3.0%	£179	1	156 d

Benchmark segmentation quality. Bootstrap Jaccard 0.75 across all three k-means clusters (well above the conventional [5] cut-off); ARI vs Ward in the moderate range; the silhouette at $k = 3$ within ~ 0.02 of $k = 2$ while delivering an extra interpretable segment.

Baseline comparison. A naive *RFM-quintile* baseline (rank customers into $5 \times 5 \times 5 = 125$ cells, group cells by hand) recovers the same A/B/C ranking on aggregate revenue but produces a coarse, manually-tuned partition with no statistical defence of cell boundaries. K-means on PCA improves on this baseline along three axes: (i) data-driven boundaries instead of hand-set quantile cuts, (ii) bootstrap-validated stability rather than analyst judgment, and (iii) a deployable scoring function that assigns new customers without re-binning.



Domain reading. **A Loyal high-value** sits above zero on every engagement feature; the segment to *protect*. **B Developing mid-value** sits above zero on tenure and frequency but at typical retail order sizes; the segment with the highest treatment-effect leverage *grow*. **C Lapsed one-time** sits below zero on every engagement feature with strong positive Recency; the segment to *triage* via controlled win-back, not blanket-targeted.

8 Conclusion

Positive results. Three actionable, validated customer segments shipped as a reproducible pipeline; the customer-share / revenue-share asymmetry recovers a finding the business can immediately operate on. **Negative results.** PC1 dominance is real and the segmentation is partially scale-dependent (mitigated by the raw-vs-scaled-PC ARI report). Ward hierarchical clustering produces two borderline clusters at the same k , so cross-method agreement is partial. **Recommendations.** Protect cluster A through retention and account-management spend; convert cluster B with measured-uplift promotions; triage cluster C with controlled win-back rather than blanket campaigns. **Caveats.** The segmentation is descriptive, not predictive; it depends on the net-vs-gross monetary choice, the winsorisation cut-off, and the one-year window. The recommended actions are testable hypotheses requiring marketing experiments to confirm.

9 Data Sources

UCI Machine Learning Repository, “Online Retail” (id 352), Chen et al. [7]: <https://archive.ics.uci.edu/dataset/352/online+retail>. Programmatic access via the `ucimlrepo` Python package (`fetch_ucirepo(id=352)`), called from `src/python/data_prep.py`. The download is cached in `data/raw/`; cleaned line items in `data/interim/`; engineered customer-level features and pipeline outputs in `data/processed/`.

10 Source Code

Repository layout (top-level): `src/python/data_prep.py` (cleaning + features), `src/R/00_setup.R ... 06_score_new_customers.R` (modelling modules), `scripts/run_pipeline.R` (orchestration), `scripts/render_reports.R` (reports), `tests/` (testthat integrity checks), `reports/` (this report and the four per-stage Rmds), `models/segmentation_pipeline.rds` (deployable artefact). **Build:** `./setup.sh` from a clean checkout creates the venv, installs Python and R deps, runs `make all`. Targets: `make data / model / reports / test / clean`. **Dependencies (open source):** Python `ucimlrepo`, `pandas`, `numpy`, `PyYAML`. R `tidyverse`, `cluster`, `factoextra`, `fpc`, `mclust`, `dendextend`, `rpart`, `rpart.plot`, `corrplot`, `GGally`, `gridExtra`, `knitr`, `rmarkdown`, `testthat`, `here`, `yaml`.

11 Bibliography

- [1] J. A. Hartigan and M. A. Wong, “Algorithm AS 136: A K-Means Clustering Algorithm,” *J. Royal Statistical Society. Series C (Applied Statistics)*, vol. 28, no. 1, pp. 100–108, 1979.
- [2] R. Tibshirani, G. Walther, and T. Hastie, “Estimating the number of clusters in a data set via the gap statistic,” *J. Royal Statistical Society. Series B*, vol. 63, no. 2, pp. 411–423, 2001.
- [3] P. J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *J. Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.
- [4] L. Hubert and P. Arabie, “Comparing partitions,” *J. Classification*, vol. 2, no. 1, pp. 193–218, 1985.
- [5] C. Hennig, “Cluster-wise assessment of cluster stability,” *Computational Statistics & Data Analysis*, vol. 52, no. 1, pp. 258–271, 2007.
- [6] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in R*, 2nd ed. New York: Springer, 2021.
- [7] D. Chen, S. L. Sain, and K. Guo, “Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining,” *J. Database Marketing & Customer Strategy Management*, vol. 19, no. 3, pp. 197–208, 2012.